

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	<i>Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.</i>		
Student Name:	T. Varun Tej	Reg. No.:	192511217
Section / Batch:		Date:	24/07/2026

Code of Conduct

I _____ T. Varun Tej/192511217 _____ (Name / Reg No)
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: T. Varun Tej

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:			100 Marks

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q1 Binary Search

5 marks

Identify the error in the following snippet:

```
int binarySearch(int arr[], int n, int key) {  
    int low = 0, high = n - 1;  
    while(low < high) {  
        int mid = (low + high) / 2;  
        if(arr[mid] == key)  
            return mid;  
        else if(arr[mid] < key)  
            low = mid;  
        else  
            high = mid;  
    }  
    return -1;  
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 int binarysearch(int arr[],int n,int key){  
3     int low = 0,high = n-1;  
4     while(low <= high){  
5         int mid = (low+high)/2;  
6         if (arr[mid] == key)  
7             return mid;  
8         else if (arr[mid]<key)  
9             low=mid+1;  
10        else  
11            high = mid-1;  
12    }  
13    return -1;  
14 }  
15 int main() {  
16     int arr[]={10,20,30,40,50};  
17     int n=5;  
18     int key =30;  
19     printf("%d",binarysearch(arr,n,key));  
20     return 0;  
21 }  
22
```

Test Input

stdin input (optional)

Output

2

Run

Save

Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q2 Stack

5 marks

Find the error in the stack deletion operation. Justify it.

```
int pop() {  
    if(top == 0) {  
        printf("Underflow");  
        return -1;  
    }  
    return stack[top--];  
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #define max 100  
3 int stack[max];  
4 int top = -1;  
5 int pop() {  
6     if (top == -1) {  
7         printf("underflow");  
8         return -1;  
9     }  
10    return stack[top--];  
11 }  
12 int main() {  
13 }  
14     stack[++top] = 10;  
15     stack[++top] = 20;  
16     stack[++top] = 30;  
17     printf("deleted element: %d", pop());  
18     return 0;  
19 }
```

Run

Save

Submit Fix

Test Input

stdin input (optional)

Output

deleted element: 30

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q3 Stack

5 marks

Check the snippet and identify the logical error.

```
int stack[5];  
int top;  
void init() {  
    top = 0;  
}
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #define max 5  
3 int stack[max];  
4 int top;  
5 void init()  
6 {  
7     top = -1;  
8 }  
9 void push(int value)  
10 { if(top==max-1)  
11 {  
12     printf("stack over\n");  
13     return;  
14 }  
15 stack[++top] = value;  
16 }  
17 int pop()  
18 {  
19     if(top == -1)  
20     {  
21         printf("stack underflow\n");  
22         return -1;  
23     }  
24     return stack[top--];  
25 }  
26 int main()  
27 {  
28     init();  
29     push(10);  
30     push(20);  
31     push(30);  
32     printf("popped element:%d\n",pop());  
33     return 0;  
34 }
```

Test Input

stdin input (optional)

Output

popped element:30

▶ Run

☁ Save

✳ Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q4 Stack

5 marks

Identify the errors in the following snippet.

```
if(ch == ')') {  
    while(stack[top] != ')') {  
        postfix[j++] = pop();  
    }  
    pop();  
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <string.h>  
3  
4 #define MAX 100  
5  
6 char stack[MAX];  
7 int top = -1;  
8 char postfix[MAX];  
9 int j = 0;  
10  
11 char pop()  
12 {  
13     if (top == -1)  
14         return '\0';  
15     return stack[top--];  
16 }  
17  
18  
19 int main()  
20 {  
21     char ch = ' ';  
22  
23     if (ch == ')')  
24     {  
25         while (top != -1 && stack[top] != '(')  
26         {  
27             postfix[j++] = pop();  
28         }  
29  
30         if (top != -1 && stack[top] == '(')  
31             pop();  
32     }  
33  
34     postfix[j] = '\0';  
35  
36     printf("Postfix: %s\n", postfix);  
37  
38     return 0;  
39 }
```

Test Input

stdin input (optional)

Output

Postfix:

Run

Save

Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q5 Stack - Balancing Symbols

5 marks

In balancing Bracket, you identify the error.

```
if((ch == '[' && stack[top] == '[') ||  
(ch == '[' && stack[top] == '[')) {  
    pop();  
} else {  
    return 0;  
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 100  
4  
5 char stack[MAX];  
6 int top = -1;  
7  
8 void push(char ch)  
9 {  
10     stack[++top] = ch;  
11 }  
12  
13 char pop()  
14 {  
15     return stack[top--];  
16 }  
17  
18 int main()  
19 {  
20     char ch = '[';  
21  
22     push('[');  
23  
24     if (top >= 0 &&  
25         ((ch == '[' && stack[top] == '[') ||  
26          (ch == '[' && stack[top] == '[')) ||  
27          (ch == '[' && stack[top] == '['))  
28     {  
29         pop();  
30         printf("Brackets are balanced\n");  
31     }  
32     else  
33     {  
34         printf("Brackets are not balanced\n");  
35     }  
36  
37     return 0;  
38 }
```

Test Input

stdin input (optional)

Output

Brackets are balanced

Run

Save

Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

Q6 Singly Linked List

5 marks

What is the issue?
temp = head
while temp.next != NULL:
print(temp.data)
temp = temp.next

★ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node
5 {
6     int data;
7     struct Node *next;
8 };
9
10 int main()
11 {
12     struct Node *head = NULL;
13     struct Node *temp;
14
15     // Creating nodes
16     head = (struct Node *)malloc(sizeof(struct Node));
17     head->data = 10;
18     head->next = NULL;
19
20     head->next = (struct Node *)malloc(sizeof(struct Node));
21     head->next->data = 20;
22     head->next->next = NULL;
23
24     head->next->next = (struct Node *)malloc(sizeof(struct Node));
25     head->next->next->data = 30;
26     head->next->next->next = NULL;
27
28     // Traversing the linked list
29     temp = head;
30
31     while (temp != NULL)
32     {
33         printf("%d ", temp->data);
34         temp = temp->next;
35     }
36
37     return 0;
38 }
```

Test Input

stdin input (optional)

Output

10 20 30

Run

Save

Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

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QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack, Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q7 Stack

5 marks

Point out the error.

```
POP(stack):  
if top == 0:  
    print("Underflow")  
else:  
    return stack[top--]
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 #define MAX 5  
4  
5 int stack[MAX];  
6 int top = -1;  
7  
8 int pop()  
9 {  
10     if (top == -1)  
11     {  
12         printf("Underflow\n");  
13         return -1;  
14     }  
15     else  
16     {  
17         return stack[top--];  
18     }  
19 }  
20  
21 int main()  
22 {  
23     stack[++top] = 10;  
24     stack[++top] = 20;  
25     stack[++top] = 30;  
26  
27     printf("Popped element: %d\n", pop());  
28  
29     return 0;  
30 }
```

Test Input

stdin input (optional)

Output

Popped element: 30

Run

Save

Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)
[← Back](#)
QUESTIONS
Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

Q8 Linked List

5 marks

Identify the errors in the following code

while current != NULL:

current.next = prev

prev = current

current = current.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 struct Node
5 {
6     int data;
7     struct Node *next;
8 };
9
10 int main()
11 {
12     struct Node *head, *current, *prev, *next;
13
14     head = (struct Node *)malloc(sizeof(struct Node));
15     head->data = 10;
16
17     head->next = (struct Node *)malloc(sizeof(struct Node));
18     head->next->data = 20;
19
20     head->next->next = (struct Node *)malloc(sizeof(struct Node));
21     head->next->next->data = 30;
22     head->next->next->next = NULL;
23
24     current = head;
25     prev = NULL;
26
27     while (current != NULL)
28     {
29         next = current->next;
30         current->next = prev;
31         prev = current;
32         current = next;
33     }
34
35     head = prev;
36
37     current = head;
38
39     printf("Reversed linked list: ");
40
41     while (current != NULL)
42     {
43         printf("%d ", current->data);
44         current = current->next;
45     }
46
47     return 0;
48 }
```

Test Input

stdin input (optional)

Output

Reversed linked list: 30 20 10

 Run

 Save

 Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack - Balancing Symbols
5 marksQ6
Singly Linked List
5 marksQ7
Stack
5 marksQ8
Linked List
5 marksQ9
Linked List
5 marksQ10
Queue
5 marks

10/10 answered

Q9 Linked List

5 marks

What is the error in the below code?

```
new.next = head  
head = new
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <stdlib.h>  
3  
4 struct Node  
5 {  
6     int data;  
7     struct Node *next;  
8 };  
9  
10 int main()  
11 {  
12     struct Node *head = NULL;  
13     struct Node *new;  
14  
15     new = (struct Node *)malloc(sizeof(struct Node));  
16  
17     new->data = 10;  
18     new->next = head;  
19     head = new;  
20  
21     printf("Linked List: ");  
22  
23     struct Node *temp = head;  
24  
25     while (temp != NULL)  
26     {  
27         printf("%d ", temp->data);  
28         temp = temp->next;  
29     }  
30  
31     return 0;  
32 }
```

Test Input

stdin input (optional)

Output

Linked List: 10

Run

Save

Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)
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QUESTIONS
Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List
5 marks

Q10
Queue
5 marks

10/10 answered

Q10 Queue

5 marks

What is missing in the below pseudocode?

```
DEQUEUE(Q):
if front == -1:
print("Underflow")
else:
return Q[front++]
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int Q[MAX],
6 int front = -1,
7 int rear = -1,
8
9 int dequeue()
10 {
11     int value;
12
13     if (front == -1)
14     {
15         printf("Underflow\n");
16         return -1;
17     }
18     else
19     {
20         value = Q[front++];
21
22         if (front > rear)
23         {
24             front = -1;
25             rear = -1;
26         }
27
28         return value;
29     }
30 }
31
32 int main()
33 {
34     Q[0] = 10;
35     Q[1] = 20;
36     Q[2] = 30;
37
38     front = 0;
39     rear = 2;
40
41     printf("Deleted element: %d\n", dequeue());
42
43     return 0;
44 }
```

Test Input

stdin input (optional)

Output

Deleted element: 10

 Run

 Save

 Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Postfix Expression
5 marks

10/10 answered

Q1 Stack

5 marks

Point out the error?

PEEK(stack);

return stack[top+1]

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int stack[MAX];
6 int top = -1;
7
8 void push(int value)
9 {
10     if (top == MAX - 1)
11     {
12         printf("Stack Overflow\n");
13         return;
14     }
15     stack[++top] = value;
16 }
17
18 int peek()
19 {
20     if (top == -1)
21     {
22         printf("Stack Underflow\n");
23         return -1;
24     }
25     return stack[top];
26 }
27
28 int main()
29 {
30     push(10);
31     push(20);
32     push(30);
33
34     printf("Top element: %d\n", peek());
35
36     return 0;
37 }
```

Test Input

stdin input (optional)

Output

Top element: 30

▶ Run

📁 Save

🚀 Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Postfix Expression
5 marks

10/10 answered

Q1 Stack

5 marks

Point out the error?

PEEK(stack);

return stack[top+1]

🔧 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2
3 #define MAX 5
4
5 int stack[MAX];
6 int top = -1;
7
8 void push(int value)
9 {
10     if (top == MAX - 1)
11     {
12         printf("Stack Overflow\n");
13         return;
14     }
15     stack[++top] = value;
16 }
17
18 int peek()
19 {
20     if (top == -1)
21     {
22         printf("Stack Underflow\n");
23         return -1;
24     }
25     return stack[top];
26 }
27
28 int main()
29 {
30     push(10);
31     push(20);
32     push(30);
33
34     printf("Top element: %d\n", peek());
35
36     return 0;
37 }
```

Test Input

stdin input (optional)



Output

Top element: 30

▶ Run

📁 Save

🚀 Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack
2 marksQ2
Circular Queue
2 marksQ3
Stack
5 marksQ4
Stack
2 marksQ5
Doubly Linked List
2 marksQ6
Queue
2 marksQ7
Queue - Deque
2 marksQ8
Binary Search
2 marksQ9
Circular Queue
5 marksQ10
Stack - Reverse Expression
2 marks

10/10 answered

qs Stack

5 marks

Point out the error.

If top == MAX:

Overflow

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdbool.h>
3
4 #define MAX 5
5
6 struct Stack {
7     int items[MAX];
8     int top;
9 };
10
11 void initialize(struct Stack* s) {
12     s->top = -1;
13 }
14
15 bool isFull(struct Stack* s) {
16     return s->top == MAX - 1;
17 }
18
19 bool isEmpty(struct Stack* s) {
20     return s->top == -1;
21 }
22
23 void push(struct Stack* s, int value) {
24     if (isFull(s)) {
25         printf("Error: Stack Overflow! Cannot push %d.\n", value);
26         return;
27     }
28     s->top++;
29     s->items[s->top] = value;
30     printf("Pushed: %d\n", value);
31 }
32
33 int pop(struct Stack* s) {
34     if (isEmpty(s)) {
35         printf("Error: Stack Underflow! Cannot pop.\n");
36         return -1;
37     }
38     int poppedValue = s->items[s->top];
39     s->top--;
40     return poppedValue;
41 }
42
43 int main() {
44     struct Stack myStack;
45     initialize(&myStack);
46
47     push(&myStack, 10);
48     push(&myStack, 20);
49     push(&myStack, 30);
50     push(&myStack, 40);
51     push(&myStack, 50);
52
53     push(&myStack, 60);
54
55     printf("Popped: %d\n", pop(&myStack));
56     printf("Popped: %d\n", pop(&myStack));
57
58     return 0;
59 }
```

Test Input

stdin input (optional)

Output

```
Pushed: 10
Pushed: 20
Pushed: 30
Pushed: 40
Pushed: 50
Error: Stack Overflow! Cannot
```

Run

Save

Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack
5 marksQ2
Circular Queue
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Doubly Linked List
5 marksQ6
Queue
5 marksQ7
Queue - Deque
5 marksQ8
Binary Search
5 marksQ9
Circular Queue
5 marksQ10
Stack - Prefix Expression
5 marks

10/10 answered

Q4 Stack

5 marks

Identify the issue in the below pseudocode.

POP():

if top == 0 → Underflow

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 #define MAX 5
5
6 struct Stack {
7     int items[MAX];
8     int top;
9 };
10
11 void initialise(struct Stack* s) {
12     s->top = -1;
13 }
14
15 bool isFull(struct Stack* s) {
16     return s->top == MAX - 1;
17 }
18
19 bool isEmpty(struct Stack* s) {
20     return s->top == -1;
21 }
22
23 void push(struct Stack* s, int value) {
24     if (isFull(s)) {
25         printf("Error: Stack Overflow!\n");
26         return;
27     }
28     s->top++;
29     s->items[s->top] = value;
30 }
31
32 int pop(struct Stack* s) {
33     if (isEmpty(s)) {
34         printf("Error: Stack Underflow!\n");
35         return -1;
36     }
37     int poppedValue = s->items[s->top];
38     s->top--;
39     return poppedValue;
40 }
41
42 int main() {
43     struct Stack myStack;
44     initialise(&myStack);
45
46     push(&myStack, 10);
47
48     printf("Popped: %d\n", pop(&myStack));
49     printf("Popped: %d\n", pop(&myStack));
50
51     return 0;
52 }
```

Test Input

stdin input (optional)

Output

Run

Save

Submit Fix

QUESTIONS

Q1
Stack
5 marksQ2
Circular Queue
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Doubly Linked List
5 marksQ6
Queue
5 marksQ7
Queue - Deque
5 marksQ8
Binary Search
5 marksQ9
Circular Queue
5 marksQ10
Stack - Non-Recursive
5 marks

10/10 answered

Q6 Queue

5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {  
    if (rear == MAX) return;  
    queue[rear++] = data;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <stdlib.h>  
3  
4 #define MAX 5  
5  
6 struct Queue {  
7     int items[MAX];  
8     int front;  
9     int rear;  
10    int size;  
11 };  
12  
13 void initialize(struct Queue* q) {  
14     q->front = 0;  
15     q->rear = 0;  
16     q->size = 0;  
17 }  
18  
19 bool isfull(struct Queue* q) {  
20     return q->size == MAX;  
21 }  
22  
23 bool isempty(struct Queue* q) {  
24     return q->size == 0;  
25 }  
26  
27 void enqueue(struct Queue* q, int data) {  
28     if (isfull(q)) {  
29         printf("Error: Queue Overflow!\n");  
30         return;  
31     }  
32     q->items[q->rear] = data;  
33     q->rear = (q->rear + 1) % MAX;  
34     q->size++;  
35     printf("enqueued: %d\n", data);  
36 }  
37  
38 int dequeue(struct Queue* q) {  
39     if (isempty(q)) {  
40         printf("Error: Queue underflow!\n");  
41         return -1;  
42     }  
43     int data = q->items[q->front];  
44     q->front = (q->front + 1) % MAX;  
45     q->size--;  
46     return data;  
47 }  
48  
49 int main() {  
50     struct Queue q;  
51     initialize(&q);  
52  
53     enqueue(&q, 10);  
54     enqueue(&q, 20);  
55     enqueue(&q, 30);  
56     enqueue(&q, 40);  
57     enqueue(&q, 50);  
58  
59     printf("enqueued: %d\n", dequeue(&q));  
60     printf("enqueued: %d\n", dequeue(&q));  
61  
62     printf("Testing reuse of space:\n");  
63     enqueue(&q, 60);  
64     enqueue(&q, 70);  
65  
66     return 0;  
67 }
```

Test Input

stdin input (optional)

Output

```
enqueued: 10  
enqueued: 20  
enqueued: 30  
enqueued: 40  
enqueued: 50  
enqueued: 10
```



Run



Save



Submit Fix

QUESTIONS

Q1
Stack
5 marksQ2
Circular Queue
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Doubly Linked List
5 marksQ6
Queue
5 marksQ7
Queue - Deque
5 marksQ8
Binary Search
5 marksQ9
Circular Queue
5 marksQ10
Stack - Morris Traversal
5 marks

10/10 answered

Q6 Queue

5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {  
    if (rear == MAX) return;  
    queue[rear++] = data;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <stdlib.h>  
3  
4 #define MAX 5  
5  
6 struct Queue {  
7     int items[MAX];  
8     int front;  
9     int rear;  
10    int size;  
11 };  
12  
13 void initialize(struct Queue* q) {  
14     q->front = 0;  
15     q->rear = 0;  
16     q->size = 0;  
17 }  
18  
19 bool isfull(struct Queue* q) {  
20     return q->size == MAX;  
21 }  
22  
23 bool isempty(struct Queue* q) {  
24     return q->size == 0;  
25 }  
26  
27 void enqueue(struct Queue* q, int data) {  
28     if (isfull(q)) {  
29         printf("Error: Queue Overflow!\n");  
30         return;  
31     }  
32     q->items[q->rear] = data;  
33     q->rear = (q->rear + 1) % MAX;  
34     q->size++;  
35     printf("enqueued: %d\n", data);  
36 }  
37  
38 int dequeue(struct Queue* q) {  
39     if (isempty(q)) {  
40         printf("Error: Queue underflow!\n");  
41         return -1;  
42     }  
43     int data = q->items[q->front];  
44     q->front = (q->front + 1) % MAX;  
45     q->size--;  
46     return data;  
47 }  
48  
49 int main() {  
50     struct Queue q;  
51     initialize(&q);  
52  
53     enqueue(&q, 10);  
54     enqueue(&q, 20);  
55     enqueue(&q, 30);  
56     enqueue(&q, 40);  
57     enqueue(&q, 50);  
58  
59     printf("enqueued: %d\n", dequeue(&q));  
60     printf("enqueued: %d\n", dequeue(&q));  
61  
62     printf("Testing reuse of space:\n");  
63     enqueue(&q, 60);  
64     enqueue(&q, 70);  
65  
66     return 0;  
67 }
```

Test Input

stdin input (optional)

Output

```
enqueued: 10  
enqueued: 20  
enqueued: 30  
enqueued: 40  
enqueued: 50  
enqueued: 60
```



Run



Save



Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1

Stack
5 marks

Q2

Circular Queue
5 marks

Q3

Stack
5 marks

Q4

Stack
5 marks

Q5

Doubly Linked List
5 marks

Q6

Queue
5 marks

Q7

Queue - Deque
5 marks

Q8

Binary Search
5 marks

Q9

Circular Queue
5 marks

Q10

Stack - Infix to Postfix
5 marks

10/10 answered

Q7 Queue - Deque

5 marks

Find the error in this Deque insertFront operation.

```
void insertFront(int data) {  
    if (front == 0) front = MAX - 1;  
    else front = front - 1;  
    deque[front] = data;  
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <stdbool.h>  
3  
4 #define MAX 5  
5  
6 struct deque {  
7     int items[MAX];  
8     int front;  
9     int rear;  
10    int size;  
11 };  
12  
13 void initialise(struct deque* dq) {  
14     dq->front = -1;  
15     dq->rear = -1;  
16     dq->size = 0;  
17 }  
18  
19 bool isfull(struct deque* dq) {  
20     return dq->size == MAX;  
21 }  
22  
23 bool isempty(struct deque* dq) {  
24     return dq->size == 0;  
25 }  
26  
27 void insertfront(struct deque* dq, int data) {  
28     if (isfull(dq)) {  
29         printf("error: deque Overflow! Cannot insert %d at front.\n", data);  
30         return;  
31     }  
32  
33     if (isempty(dq)) {  
34         dq->front = 0;  
35         dq->rear = 0;  
36     } else if (dq->front == 0) {  
37         dq->front = MAX - 1;  
38     } else {  
39         dq->front = dq->front - 1;  
40     }  
41  
42     dq->items[dq->front] = data;  
43     dq->size++;  
44     printf("inserted at front: %d\n", data);  
45 }  
46  
47 void insertrear(struct deque* dq, int data) {  
48     if (isfull(dq)) {  
49         printf("error: deque Overflow!\n");  
50         return;  
51     }  
52  
53     if (isempty(dq)) {  
54         dq->front = 0;  
55         dq->rear = 0;  
56     } else if (dq->rear == MAX - 1) {  
57         dq->rear = 0;  
58     } else {  
59         dq->rear = dq->rear + 1;  
60     }  
61  
62     dq->items[dq->rear] = data;  
63     dq->size++;  
64 }  
65  
66 int main() {  
67     struct deque dq;  
68     initialise(&dq);  
69 }
```

Test Input

Write input (optional)

Output

```
inserted at front: 10  
inserted at front: 20  
inserted at front: 30  
--- testing Overflow Condition ---
```


CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack
5 marksQ2
Circular Queue
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Doubly Linked List
5 marksQ6
Queue
5 marksQ7
Queue - Deque
5 marksQ8
Binary Search
5 marksQ9
Circular Queue
5 marksQ10
Stack - Postfix Expression
5 marks

10/10 answered

Q8 Binary Search

5 marks

Identify the bug in this Recursive Binary Search.

```
int binarySearch(int arr[], int l, int r, int x) {  
    int mid = l + (r - l) / 2;  
    if (arr[mid] == x) return mid;  
    if (arr[mid] > x) return binarySearch(arr, l, mid, x);  
    return binarySearch(arr, mid, r, x);  
}
```

★ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2  
3 int binarySearch(int arr[], int l, int r, int x) {  
4     if (l > r) {  
5         return -1;  
6     }  
7  
8     int mid = l + (r - l) / 2;  
9  
10    if (arr[mid] == x) {  
11        return mid;  
12    }  
13  
14    if (arr[mid] > x) {  
15        return binarySearch(arr, l, mid - 1, x);  
16    }  
17  
18    return binarySearch(arr, mid + 1, r, x);  
19 }  
20  
21 int main() {  
22     int arr[] = {2, 3, 4, 10, 40};  
23     int n = sizeof(arr) / sizeof(arr[0]);  
24  
25     int target1 = 10;  
26     int result1 = binarySearch(arr, 0, n - 1, target1);  
27     printf("Searching for %d... Found at index: %d\n", target1, result1);  
28  
29     int target2 = 5;  
30     int result2 = binarySearch(arr, 0, n - 1, target2);  
31     printf("Searching for %d... Found at index: %d\n", target2, result2);  
32  
33     return 0;  
34 }
```

Test Input

stdin input (optional)

Output

```
Searching for 10... Found at index:  
3  
Searching for 5... Found at index:  
-1
```

Run

Save

Submit Fix

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue Dequeue
5 marks

Q8
Binary Search
5 marks

Q9
Circular Queue
5 marks

Q10
Stack Postfix Expression
5 marks

10/10 answered

Q5 Circular Queue

5 marks

Identify the bug in this Circular Queue dequeue logic.

```
int dequeue() {
    if (front == -1) return -1;
    int data = queue[front];
    front = (front + 1) % MAX;
    return data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>
2 #include <stdbool.h>
3
4 #define MAX 5
5
6 struct CircularQueue {
7     int items[MAX];
8     int front;
9     int rear;
10 };
11
12 void initialise(struct CircularQueue* q) {
13     q->front = -1;
14     q->rear = -1;
15 }
16
17 bool isFull(struct CircularQueue* q) {
18     return ((q->rear + 1) % MAX) == q->front;
19 }
20
21 bool isEmpty(struct CircularQueue* q) {
22     return q->front == -1;
23 }
24
25 void enqueue(struct CircularQueue* q, int value) {
26     if (isFull(q)) {
27         printf("Error: Queue Overflow!\n");
28         return;
29     }
30     if (isEmpty(q)) {
31         q->front = 0;
32     }
33     q->rear = (q->rear + 1) % MAX;
34     q->items[q->rear] = value;
35     printf("Enqueued: %d\n", value);
36 }
37
38 int dequeue(struct CircularQueue* q) {
39     if (isEmpty(q)) {
40         printf("Error: Queue Underflow!\n");
41         return -1;
42     }
```

Test Input

stdin input (optional)

Output

```
Enqueued: 10
Enqueued: 20
Popped: 10
Popped: 20
Testing enqueue after
reaching empty queue
```

CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 2)

← Back

QUESTIONS

Q1
Stack
5 marksQ2
Circular Queue
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Doubly Linked List
5 marksQ6
Queue
5 marksQ7
Queue - Deque
5 marksQ8
Binary Search
5 marksQ9
Circular Queue
5 marksQ10
Stack - Postfix Expression
5 marks

10/10 answered

Q10 Stack - Postfix Expression

5 marks

Find the bug in the evaluation of a Postfix expression.

```
while (expression[i] != '\0') {  
    if (isdigit(expression[i])) push(expression[i] - '0');  
    else {  
        int op1 = pop();  
        int op2 = pop();  
        // perform operation: result = op1 + op2;  
        push(result);  
    }  
}
```

🔍 Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #include <stdio.h>  
2 #include <ctype.h>  
3 #include <stdlib.h>  
4  
5 #define MAX 100  
6  
7 struct Stack {  
8     int items[MAX];  
9     int top;  
10 };  
11  
12 void initialize(struct Stack* s) {  
13     s->top = -1;  
14 }  
15  
16 void push(struct Stack* s, int value) {  
17     if (s->top == MAX - 1) return;  
18     s->items[++(s->top)] = value;  
19 }  
20  
21 int pop(struct Stack* s) {  
22     if (s->top == -1) return 0;  
23     return s->items[(s->top)--];  
24 }  
25  
26 int evaluatePostfix(char* expression) {  
27     struct Stack s;  
28     initialize(&s);  
29     int i = 0;  
30  
31     while (expression[i] != '\0') {  
32         if (isdigit(expression[i])) {  
33             push(&s, expression[i] - '0');  
34         } else {  
35             int op2 = pop(&s);  
36             int op1 = pop(&s);  
37             int result = 0;  
38
```

Test Input

stdin input (optional)

Output

Postfix Expression: 231*+9-
Evaluation Result: -4